Criterion B: Record of Tasks

Table 2.6 below chronologically records every task undertaken during the planning, design, development and evaluation of FocusGrid, alongside the time spent, the completion date, and the IB Computer Science criterion (A, B, C, D or E) to which the task contributes.

No.	Task Action	Task Outcome	Time	Completion Date	Crit.
1	Client Interview 1	Identified client's requirements and created an initial Criteria for Success (CFS).	0.5h	5 May 2025	**A**
2	Discussion with advisor on implementation	Confirmed Flutter, Hive as the final technical product	1h	5 May 2025	**A**
3	Final input from advisor on solution rationale	Advisor reviewed and approved the proposed solution for each client concern.	0.5h	7 May 2025	**A**
4	Completion of Criterion A (Planning)	Problem definition, solution rationale and CFS finalised and written up	1.5h	12 May 2025	**A**
5	Sketching of GUI screens	Hand-drawn sketches of all main screens produced, covering every item in the CFS	1h	15 May 2025	**B**
6	Client Interview 2	Client reviewed the sketches and gave feedback on layout and missing features.	0.5h	18 May 2025	**B**
7	Revised digital mock-up of GUI	Digital mock-up screens created in the dark neon theme based on client feedback.	2h	22 May 2025	**B**
8	Labelling of UI elements	Every widget on the mock-ups labelled with its corresponding Flutter widget, to aid Criterion C.	1h	24 May 2025	**B**
9	Creation of system flowchart	Shows navigation between the navigation bar and all screens.	1.5h	27 May 2025	**B**
10	Creation of screen-level flowcharts and pseudocode	Flowchart and pseudocode produced for each screen to guide coding logic.	8h	5 June 2025	**B**
11	Planning of data structures	Hive box structure planned, including model fields and data types.	1h	6 June 2025	**B**
12	Creation of Data Flow Diagram (DFD)	Shows the flow of data between the screens, Hive storage and the Google Calendar API.	1h	7 June 2025	**B**
13	Creation of UML class diagram	Contains all planned classes, attributes and methods used in development.	2h	10 June 2025	**B**
14	Creation of test plan	Test cases drafted to verify every CFS item once the app is built.	1h	12 June 2025	**B**
15	Completion of Criterion B (Design)	All design artefacts reviewed with the advisor and finalised.	1h	13 June 2025	**B**
16	Set up Integrated Development Environment (IDE)	Flutter SDK, VS Code and Android/iOS emulators installed and configured.	2h	20 June 2025	**C**
17	Built Reminder data model (CFS 2a)	Reminder class created with Hive type annotations; type adapter generated.	1.5h	23 June 2025	**C**
18	Implemented HiveHelper CRUD methods	Add, get, update and delete methods verified with sample data.	3h	27 June 2025	**C**
19	Built navigation bar and app shell (main.dart)	BottomNavigationBar routes between screens; Home screen shown by default.	2h	30 June 2025	**C**
20	Built Home screen calendar (CFS 3a–3c)	TableCalendar integrated; tapping a date filters the reminder list beneath it.	5h	8 July 2025	**C**

No.	Task Action	Task Outcome	Time	Completion Date	Crit.
21	Built Add Reminder screen UI (CFS 2a)	Form fields for all six reminder attributes laid out to match the mock-up.	4h	15 July 2025	**C**
22	Implemented reminder validation and save logic (CFS 2c)	Mandatory fields validated before save; reminder written to Hive on success.	3h	18 July 2025	**C**
23	Built Reminder List screen (CFS 4a–4c)	Reminders sorted chronologically; tap opens detail; edit and delete supported.	4h	24 July 2025	**C**
24	Fixed pixel-overflow bug on Add Reminder screen	Form wrapped in SingleChildScrollView, removing the overflow error.	1h	24 July 2025	**C**
25	Implemented NotificationService (CFS 7a–7b)	Notifications scheduled at the exact reminder time using flutter_local_notifications.	3h	30 July 2025	**C**
26	Unit-tested HiveHelper and NotificationService	Verified CRUD accuracy and confirmed notifications fire within 60 seconds of schedule.	2h	16 August 2025	**C**
27	Executed full test plan	Ran every test case from the Criterion B test plan against the finished app.	1.5h	20 August 2025	**C**
28	Fixed bugs found during testing	Minor colour-marker rendering bug on the calendar corrected.	1h	21 August 2025	**C**
29	Documented techniques used	Explained and justified each technique used: Hive, OAuth, notifications, free-slot detection.	2h	25 August 2025	**C**
30	Documented data structures and Flutter library imports	Described Hive boxes and models, and every imported Flutter package.	2.5h	28 August 2025	**C**
31	Documented code structure	Explained the purpose of every Dart file in the Visual Studio Code project.	2h	1 September 2025	**C**
32	Completion of Criterion C (Development)	All code, screenshots and documentation compiled and finalised.	3h	4 September 2025	**C**
33	Application installed on client's mobile device	FocusGrid installed on Escher's Android phone for real-world use.	0.5h	8 September 2025	**E**
34	Client Interview 3	Client explored the app independently; feedback checked against every CFS item.	1h	10 September 2025	**E**
35	Consultation with advisor on recommendations for improvement	Discussed improvements, including bi-directional sync and custom themes.	0.5h	11 September 2025	**E**
36	Completion of Criterion E (Evaluation)	Evaluation against the CFS, strengths, weaknesses and improvements written up.	2h	15 September 2025	**E**
37	Completion of Criterion D (Video)	Video recorded and edited, demonstrating every CFS item in turn.	6h	20 September 2025	**D**

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